

Support vector machines I: maximum margin

The widest street between two classes: the margin $2/\|w\|$, the primal QP, its dual, support vectors and prediction.

THE QUESTION

Many hyperplanes separate two classes. Which one should we pick, and how do we compute it?

THE INTUITION

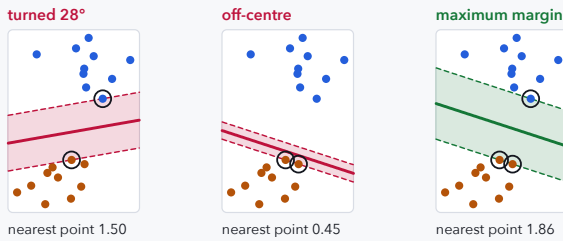
Pick the line with the widest empty street around it. Scale w so the nearest points give $w^T x + b = \pm 1$; the street is then $2/\|w\|$ wide, so widening it means shrinking $\|w\|$. Only the points on the edges of the street decide the answer.

KEY TAKEAWAY

Minimise $\frac{1}{2}\|w\|^2$ subject to $y_i(w^T x_i + b) \geq 1$, a QP. Its dual gives $w = \sum_i \alpha_i y_i x_i$ with $\alpha_i > 0$ only on the support vectors. Predict $\text{sign}(w^T x + b)$.

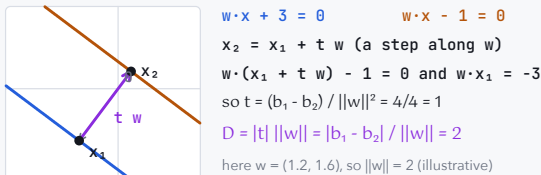
$y_i = \pm 1 \rightarrow \text{width } 2/\|w\| \rightarrow \min \frac{1}{2}\|w\|^2, y_i(w \cdot x_i + b) \geq 1 \rightarrow \text{dual in } \alpha \rightarrow \alpha_i > 0 \text{ on support vectors} \rightarrow \text{sign}(w \cdot x + b)$

1 Many lines separate; take the widest street



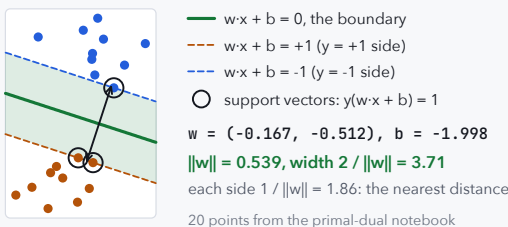
Labels are $y_i \in \{-1, +1\}$. Separable data admit infinitely many hyperplanes $w^T x + b = 0$: a point in 1D, a line in 2D, a plane in 3D. On the primal-dual notebook's 20 points, a line hugging one class keeps 0.45 to its nearest point and a tilted one 1.50. The maximum-margin line keeps 1.86 on both sides.

2 Parallel planes are $|b_1 - b_2| / \|w\|$ apart



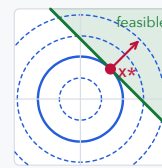
w is perpendicular to $w^T x + b = 0$: for two points on it, $w^T (x - x_0) = 0$. Step from x_1 on the first plane along w , $x_2 = x_1 + t w$, and put x_2 into the second plane: $t = (b_1 - b_2) / \|w\|^2$, so the distance is $D = |t| \|w\| = |b_1 - b_2| / \|w\|$.

3 Fix the scale and the width is $2/\|w\|$



Scaling w and b together leaves the plane where it is, so choose the scale that puts the nearest points on $w^T x + b = \pm 1$. These two planes have $b_1 - b_2 = 2$: the street is $2/\|w\|$ wide. Every point must lie on or beyond its own plane: $y_i(w^T x_i + b) \geq 1$.

4 The primal is a quadratic program

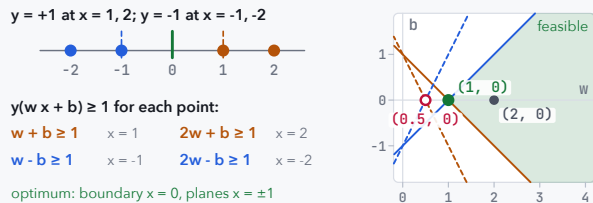


Maximising $2/\|w\|$ is minimising $\|w\| = \sqrt{w^T w}$; squaring and halving make it smooth:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad y_i(w^T x_i + b) \geq 1, \quad i = 1, \dots, N.$$

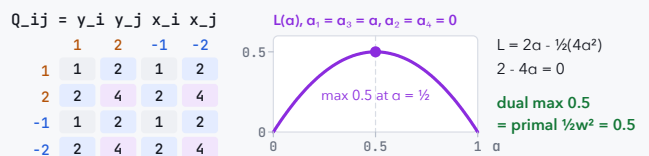
A quadratic objective under linear constraints is a QP: convex, with one minimum, and standard solvers handle it.

5 The 1D example: $w = 1, b = 0$



Four points give four constraints on (w, b) . Those from $x = \pm 1$ give $w \geq 1 + |b|$ (shaded) and imply the other two. $\frac{1}{2}w^2$ wants w small, so the optimum is the corner $(1, 0)$: boundary $x = 0$, width 2. $(2, 0)$ is feasible with width 1. $(0.5, 0)$ separates the points but breaks the constraints at $x = \pm 1$.

6 Lagrange multipliers give a dual in α



One $\alpha_i \geq 0$ per constraint (the constrained optimisation and KKT sheet): $L = \frac{1}{2}\|w\|^2 - \sum_i \alpha_i [y_i(w^T x_i + b) - 1]$. $\partial L / \partial w = 0$ gives $w = \sum_i \alpha_i y_i x_i$, $\partial L / \partial b = 0$ gives $\sum_i \alpha_i y_i = 0$, and substituting leaves

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j, \quad \alpha_i \geq 0, \quad \sum_i \alpha_i y_i = 0.$$

In 1D only $x = \pm 1$ are active (panel 7); symmetry sets $\alpha_1 = \alpha_3 = \alpha$.

Support vectors, prediction and logistic regression

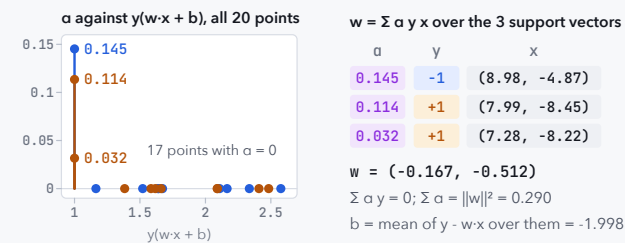
Complementary slackness, the solved dual in 1D and 2D, prediction in both forms, and what moves the boundary.

7 Only points on the margin get $\alpha_i > 0$

x	1	2	-1	-2	on the margin: $yf = 1$, α can be > 0
y	+1	+1	-1	-1	
y f(x)	1	2	1	2	off the margin: $yf > 1$ forces $\alpha = 0$
α	0.5	0	0.5	0	
$\alpha (y f(x) - 1)$	0	0	0	0	every product is 0

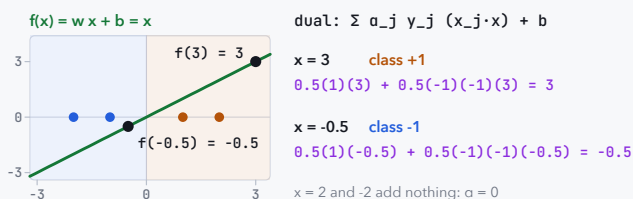
At the optimum, complementary slackness holds: $\alpha_i [y_i (w^T x_i + b) - 1] = 0$ for every i . A point beyond its plane has $\alpha_i = 0$; a point with $\alpha_i > 0$ lies on $w^T x + b = \pm 1$. These are the support vectors. (A point on the plane may still have $\alpha_i = 0$; slackness only rules out $\alpha_i > 0$ off it.) In 1D, $w = \frac{1}{2}(1)(1) + \frac{1}{2}(-1)(-1) = 1$. Multiplying $y_i (w x_i + b) = 1$ by y_i ($y_i^2 = 1$) gives $b = y_i - w x_i = 0$ at either support vector; in practice average b over all of them.

8 The 2D example has three support vectors



The primal-dual notebook's 20 points ($C = 10^6$, so practically hard margin), solved to KKT precision: 17 points get $\alpha_i = 0$. At the optimum $\sum_i \alpha_i = \|w\|^2$, and primal and dual objectives are both 0.145. A QP solver takes $Q_{ij} = y_i y_j x_i^T x_j$; minimise $\frac{1}{2} \alpha^T Q \alpha - 1^T \alpha$ with $\alpha \geq 0$, $y^T \alpha = 0$ (the cvxopt notebook). Its 40 points give two support vectors, $\alpha = 0.0543$ each, and 0.0543 for both objectives.

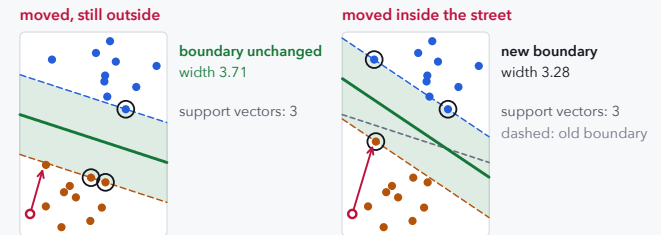
9 Predict with the sign, in either form



$$\hat{y}(x) = \text{sign}(w^T x + b) = \text{sign}\left(\sum_{j \in \text{SV}} \alpha_j y_j x_j^T x + b\right).$$

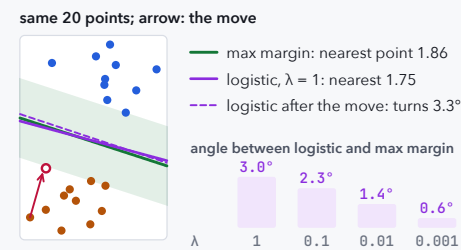
Both give +1 at $x = 3$ and -1 at $x = -0.5$. The dual form needs only the support vectors and a dot product with each: $O(N_{\text{SV}} D)$ work per query, where D is the number of features. It is also the form a kernel plugs into.

10 Only the support vectors fix the boundary



Move a point with $\alpha_i = 0$ anywhere outside the street and nothing changes: w and b agree to 10^{-9} . Move it inside and it becomes a support vector: the boundary turns and the width falls from 3.71 to 3.28. Dropping every non-support point gives the same classifier.

11 Logistic regression listens to every point



Both predict with $\text{sign}(w^T x + b)$. Logistic regression (the logistic regression sheet) minimises cross-entropy over all points and gives probabilities; on separable data it needs a penalty $\lambda \|w\|^2$ to stay finite. With $\lambda = 1$ its boundary is 3.0° off the maximum-margin one, nearest point 1.75 against 1.86. As $\lambda \rightarrow 0$ it turns towards it. The move of panel 10 turns it 3.3°; the SVM stays put.

12 Pause and think

Planes $w^T x + 3 = 0$ and $w^T x - 1 = 0$, $\|w\| = 2$?
 $D = 4/2 = 2$.

Why the dual? Only $x_i^T x_j$ appears, so a kernel can replace it.

Predict $x = -0.5$?
 $\text{sign}(-0.5) = -1$ in both forms.

Why minimise $\frac{1}{2} \|w\|^2$?
 Same optimum as maximising $2/\|w\|$, and it is a QP.

Is $w = 1$ the only 1D separator? No, but it has the smallest feasible $|w|$. Support vectors in 1D? $x = \pm 1$, each with $\alpha = \frac{1}{2}$.

Classes overlap? No (w, b) is feasible: slack variables and C , next sheet.
 SVM or logistic regression? Logistic for probabilities, SVM for a sparse boundary.

$$\text{width } 2/\|w\| \cdot \min \frac{1}{2} \|w\|^2 \text{ s.t. } y_i (w^T x_i + b) \geq 1 \cdot w = \sum_i \alpha_i y_i x_i \cdot \alpha_i > 0 \text{ only on support vectors} \cdot \text{sign}(w^T x + b)$$

Before: the constrained optimisation and KKT sheet; next: the support vector machines II sheet (kernels and soft margin). Source: lecture slides "Support vector machines" with their handwritten notes, and the SVM, primal-dual and cvxopt notebooks (nipunbatra.github.io/ml-teaching); the 20 points are the primal-dual notebook's, solved here with NumPy (SMO on the dual, checked against KKT). Corrected: the notes label the $x = -2$ constraint $2w + b = 1$ (it is $2w - b \geq 1$), and a quiz calls $w = 0.5, b = 0$ another solution (it breaks the constraints).